FINAL YEAR PROJECT

Smart Wi-Fi Network Intelligence and Dead-Zone Detection System

Project Proposal & Technical Overview

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1. Project Description

Smart Wi-Fi Network Intelligence and Dead-Zone Detection System is an intelligent system designed to analyze Wi-Fi network performance and identify weak-signal or dead-zone areas. It monitors signal strength, network speed, connectivity, and coverage patterns to detect performance issues. The system provides users with network insights, identifies locations with poor connectivity, and suggests possible improvements for better Wi-Fi coverage. It can help homes, offices, universities, and organizations maintain reliable, efficient, and stable wireless networks.

2. Key Objectives

- Monitor and analyze Wi-Fi signal strength and network performance.
- Identify areas with weak connectivity and Wi-Fi dead zones.
- Present network information through a clear and user-friendly dashboard.
- Use data analysis and machine learning techniques to identify network patterns.
- Provide useful insights for improving Wi-Fi coverage and reliability.

3. Technologies Used

- **Python** – Core programming, network monitoring, and analysis.
- **Flask** – Backend and web application framework.
- **HTML5 & CSS3** – Frontend structure and responsive interface.
- **JavaScript** – Interactive dashboard and data handling.
- **Wi-Fi / Network APIs** – Collecting signal and network information.
- **NumPy & Pandas** – Data processing and analysis.
- **Scikit-learn** – Machine learning and dead-zone detection.

- **SQLite / MySQL** – Database management and storage.
- **Chart.js** – Network performance graphs and visualization.
- **Git & GitHub** – Version control and project management.

4. Expected Outcome

The proposed system will provide an easy-to-use platform for understanding Wi-Fi coverage and network performance. By detecting weak-signal areas and presenting meaningful network insights, the system can support better placement of access points and improve overall wireless connectivity.